

LINUX COMMANDS ASSIGNMENT

Submitted By

Name: Muhammad Adnan

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Task 1: Creating and Renaming Files/Directories

Command 1- mkdir test_dir

Command 2- cd test-dir

Command 3- touch example.txt

Command 4- mv example.txt renamed_example.txt

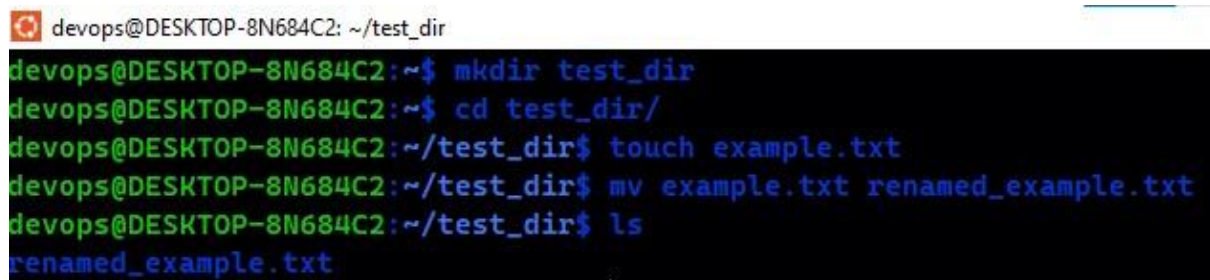
Explanation

Created a directory named test_dir using mkdir.

Get inside the directory using cd command.

Created an empty text file named example.txt using touch command.

Renamed the example.txt to renamed_example.txt using mv command.



```
devops@DESKTOP-8N684C2: ~/test_dir
devops@DESKTOP-8N684C2:~$ mkdir test_dir
devops@DESKTOP-8N684C2:~$ cd test_dir/
devops@DESKTOP-8N684C2:~/test_dir$ touch example.txt
devops@DESKTOP-8N684C2:~/test_dir$ mv example.txt renamed_example.txt
devops@DESKTOP-8N684C2:~/test_dir$ ls
renamed_example.txt
```

Task 2: Viewing File Contents

Command 1- cat /etc/passwd

Command 2- head -5 /etc/passwd

Command 3- tail -5 /etc/passwd

Explanation

Viewed the content of /etc/passwd using cat command.

Viewed only first 5 lines of /etc/passwd using head command.

Viewed only last 5 lines of /etc/passwd using tail command.

```
devops@DESKTOP-8N684C2: ~  
devops@DESKTOP-8N684C2:~$ cat /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin  
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin  
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin  
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin  
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin  
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin  
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin  
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin  
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false  
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin  
syslog:x:102:102::/nonexistent:/usr/sbin/nologin  
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin  
uidd:x:103:103::/run/uidd:/usr/sbin/nologin  
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin  
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin  
adnan:x:1000:1000:,,,:/home/adnan:/bin/bash  
random:x:1001:1001:,,,:/home/random:/bin/bash  
sshd:x:105:65534:/:run/sshd:/usr/sbin/nologin  
devops:x:1004:1004:,,,:/home/devops:/bin/bash
```

```
devops@DESKTOP-8N684C2:~$ head -5 /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
devops@DESKTOP-8N684C2:~$ tail -5 /etc/passwd  
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin  
adnan:x:1000:1000:,,,:/home/adnan:/bin/bash  
random:x:1001:1001:,,,:/home/random:/bin/bash  
sshd:x:105:65534:/:run/sshd:/usr/sbin/nologin  
devops:x:1004:1004:,,,:/home/devops:/bin/bash
```

Task 3: Searching the Patterns

Command 1- `cat /etc/passwd | grep root`

Explanation-

Viewing the content of the `/etc/passwd` and after getting output of this command searching the string “**root**” in it.

```
devops@DESKTOP-8N684C2: ~  
devops@DESKTOP-8N684C2:~$ cat /etc/passwd | grep root  
root:x:0:0:root:/root:/bin/bash
```

Task 4: Zipping and Unzipping

Command 1- `zip -r test_dir.zip test_dir`

Command 2- `mkdir unzipped_dir`

Command 3- `cd unzipped_dir`

Command 4- `unzip ../test_dir.zip -d .`

Explanation

Zipped the `test_dir` recursively using `zip` command.

Created a directory named `unzipped_dir`.

Get inside the `unzipped_dir` using `cd`.

Unzipped the `test_dir.zip` using `unzip` command since we are in `unzipped_dir` and our zipped file is outside of the directory and it is in the previous directory so we used `../test_dir.zip` and we use `-d` to specify the directory and use `.` to mention that unzipped file will be in the current directory .

```
devops@DESKTOP-8N684C2: ~/unzipped_dir  
devops@DESKTOP-8N684C2:~$ ls  
compose-project  devops  docker-networking  docker-volume-demo  feedback-project.txt  gitlearning  projects  ssh  test_dir  
devops@DESKTOP-8N684C2:~$ zip -r test_dir.zip test_dir/  
  adding: test_dir/ (stored 0%)  
  adding: test_dir/renamed_example.txt (stored 0%)  
devops@DESKTOP-8N684C2:~$ ls  
compose-project  devops  docker-networking  docker-volume-demo  feedback-project.txt  gitlearning  projects  ssh  test_dir  test_dir.zip  
devops@DESKTOP-8N684C2:~$ mkdir unzipped_dir  
devops@DESKTOP-8N684C2:~$ cd unzipped_dir/  
devops@DESKTOP-8N684C2:~/unzipped_dir$ unzip ../test_dir.zip -d .  
Archive:  ../test_dir.zip  
   creating: ./test_dir/  
  extracting: ./test_dir/renamed_example.txt  
devops@DESKTOP-8N684C2:~/unzipped_dir$ ls  
test_dir
```

Task 5: Downloading Files

Command 1- `wget https://example.com/sample.txt`

Explanation

Downloading the file sample.txt using the given link and wget command after typing the command getting the 404 ERROR because the file does not exist on the specified url or server.

```
devops@DESKTOP-8N684C2: ~  
devops@DESKTOP-8N684C2:~$ wget https://example.com/sample.txt  
--2026-07-20 06:20:28-- https://example.com/sample.txt  
Resolving example.com (example.com)... 172.66.147.243, 104.20.23.154, 2606:4700:8395:72db:f239:9b4:ef6b:ff98  
Connecting to example.com (example.com)[172.66.147.243]:443... connected.  
HTTP request sent, awaiting response... 404 Not Found  
2026-07-20 06:20:28 ERROR 404: Not Found.  
  
devops@DESKTOP-8N684C2:~$ ls  
compose-project  docker-networking  feedback-project.txt  projects  test_dir  unzipped_dir  
devops          docker-volume-demo  gitlearning           ssh       test_dir.zip
```

Task 6: Changing Permissions

Command 1- `touch secure.txt`

Command 2- `chmod 444 secure.txt`

Command 3- `ls -l`

Explanation

Created an empty file named secure.txt using touch command.

Gives read-only permission to the users,groups.others using chmod 444 command as 4 represent the read,3 represent the write and 1 represent the execute.

Checking the permission using ls -l command as it gives the long list and necessary information about the files.

```
devops@DESKTOP-8N684C2: ~  
devops@DESKTOP-8N684C2:~$ touch secure.txt  
devops@DESKTOP-8N684C2:~$ ls  
compose-project  docker-networking  feedback-project.txt  projects  ssh  test_dir.zip  
devops          docker-volume-demo  gitlearning           secure.txt  test_dir  unzipped_dir  
devops@DESKTOP-8N684C2:~$ chmod 444 secure.txt  
devops@DESKTOP-8N684C2:~$ ls -l  
total 2032  
drwxrwxr-x 4 devops devops 4096 Jun  2 20:03 compose-project  
drwxrwxr-x 2 devops devops 4096 Jun  1 20:17 devops  
drwxrwxr-x 2 devops devops 4096 Jun  2 18:25 docker-networking  
drwxrwxr-x 2 devops devops 4096 Jun  2 18:00 docker-volume-demo  
-rw-rw-r-- 1 devops devops 2035712 Jun 18 19:42 feedback-project.txt  
drwxrwxr-x 6 devops devops 4096 Jun  4 19:53 gitlearning  
drwxrwxr-x 3 devops devops 4096 Jun 18 17:30 projects  
-r--r--r-- 1 devops devops 0 Jul 20 06:24 secure.txt  
drwxrwxr-x 2 devops devops 4096 Jun 19 18:04 ssh  
drwxrwxr-x 2 devops devops 4096 Jul 20 06:11 test_dir  
-rw-rw-r-- 1 devops devops 352 Jul 20 06:17 test_dir.zip  
drwxrwxr-x 3 devops devops 4096 Jul 20 06:18 unzipped_dir
```

Task 7: Working with Environmental variables

Command 1- export MY_VAR="Hello, Linux!"

Command 2- echo \$MY_VAR

Command 3- printenv .

Explanation

Exporting my Environmental variable named MY_VAR="Hello, Linux!" using export command.

Checking the env variable using echo command .

Verifying the environmental variable using printenv command.

```
devops@DESKTOP-8N684C2:~$ export MY_VAR="Hello, Linux!".
devops@DESKTOP-8N684C2:~$ echo $MY_VAR
Hello, Linux!.
devops@DESKTOP-8N684C2:~$ printenv
SHELL=/bin/bash
PWD=/home/devops
LOGNAME=devops
XDG_SESSION_TYPE=tty
MOTD_SHOWN=update-motd
HOME=/home/devops
LANG=C.UTF-8
MY_VAR=Hello, Linux!.
```